Advanced Computer Networks

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MAC Protocols: CSMA Basics

DSSRG: Decentralized Smart Systems Research Group

https://sites.google.com/iitbbs.ac.in/dssrg
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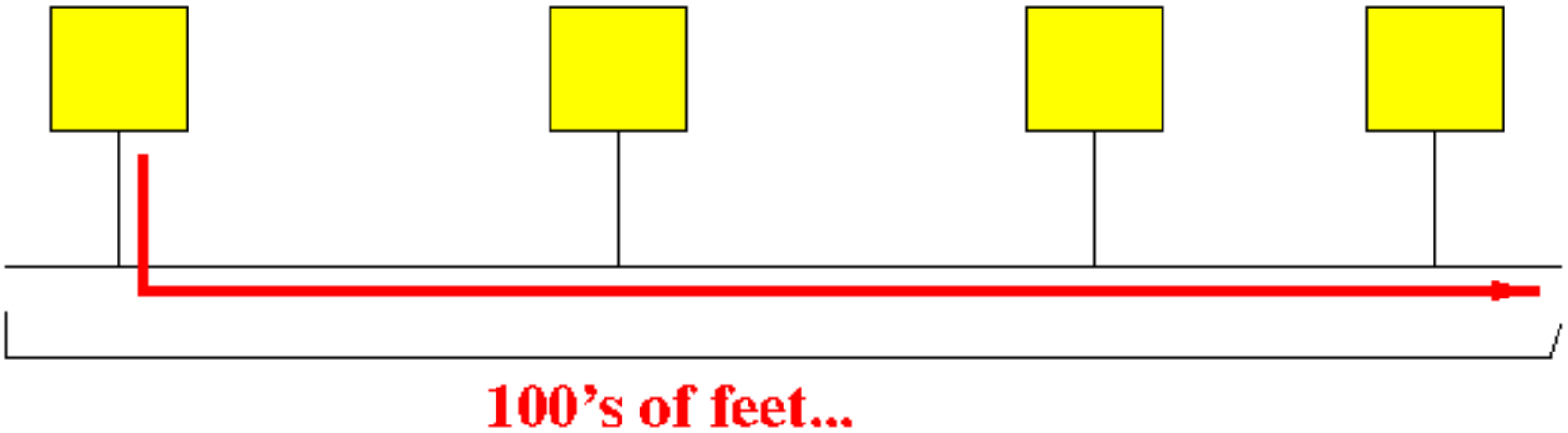
CSMA: the Carrier Sense Multiple Access protocol

- **Local Area Network (LAN)**
- **Distances:**
- **Local Area Networks** covers an **office building**
- Distances between **any 2 nodes** in a **LAN ≤ 500 m**

- **Consequence:**
- The signal will take **≤ 500/200,000,000 = 2.5 × 10^{-6} sec** to **reach** any other node in the network.

- When a node starts a transmission, the **start of the transmission** will reach **all other nodes** in the LAN **before the whole message is transmitted**

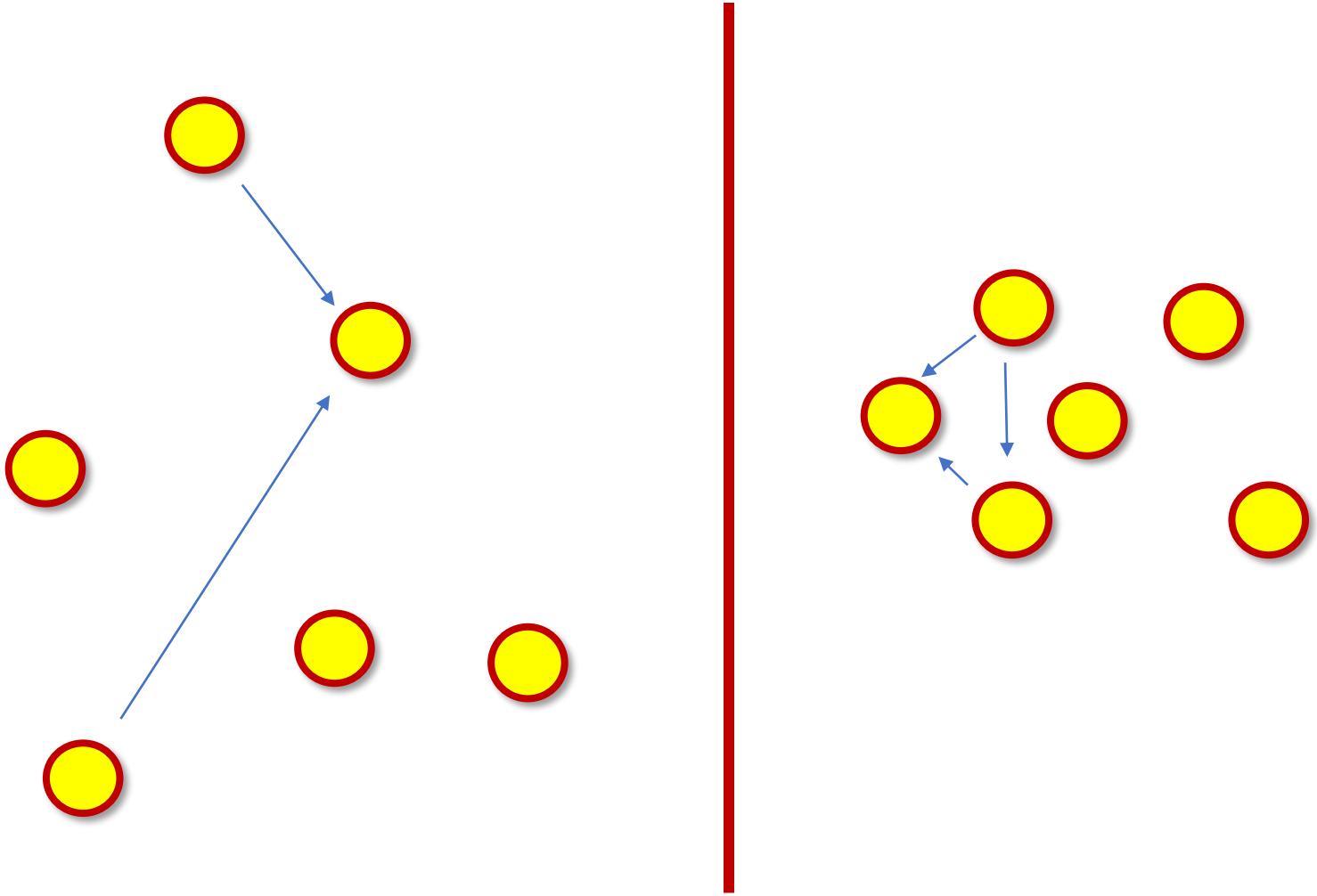
Example:



Conclusion

We can **reduce the probability of collision** by **sensing the channel (transmission medium)** ***before*** **transmitting a message**

The differences in the two settings



Carrier Sense Multiple Access (CSMA) protocol

- **Slotted and unslotted:**
- **CSMA protocol** can be **slotted** or **unslotted**
- **Unslotted: transmission** can **begin any time (unrestraint)**
- **Slotted: Transmissions** can **only begin** at the **start of a slot**

The *general* CSMA protocol:

- When a node has a **message** to transmit, it **first *senses* the broadcast channel** (or **carrier**) to determine if it is **idle** (not used by some other node)
- If the **carrier** is **busy**, the **node** will ***wait* and retry at a later time**
- If the carrier is **idle**, the node will ***attempt* to transmit**
- ***How*** a node decides to **wait** and **retry** will be explain below...

Note:

- The name **Carrier Sense Multiple Access** should be clear now...
- The protocol calls for **sensing the carrier (channel)**
- And it needs to **try many times** before an attempt will be **successful** (= **multiple access**)

Why did Aloha not use carrier sensing ?

- So why can't **Aloha** use **channel sensing** ?

- **Purpose of sensing the channel**

- The **purpose** of **sensing the channel** is to **reduce the *likelihood*** of **collisions**

Reason that CSMA use *channel sensing*

- **Sensing the channel will reduce** the **likelihood of a collision**
- **Because:**
- The **length (in time) of a packet** is **much longer** that the **propagation delay** from the sender to the sensor

- So:
- If the **channel is idle**, the **chance is very good** (in a LAN) that **nobody is presently transmitting**.
- If the **channel is busy**, the **chance is very good** (in a LAN) that **somebody is presently transmitting**.

Examples that show carrier sensing in Aloha is useless

- Suppose a node senses the channel is ***idle*** **(no transmission)**
- **Question:**
- Can this **fact** help a node **reduce the likelihood** of a **collision** ?
- **NO !!!**
- A node can get a **false channel idle** indication:

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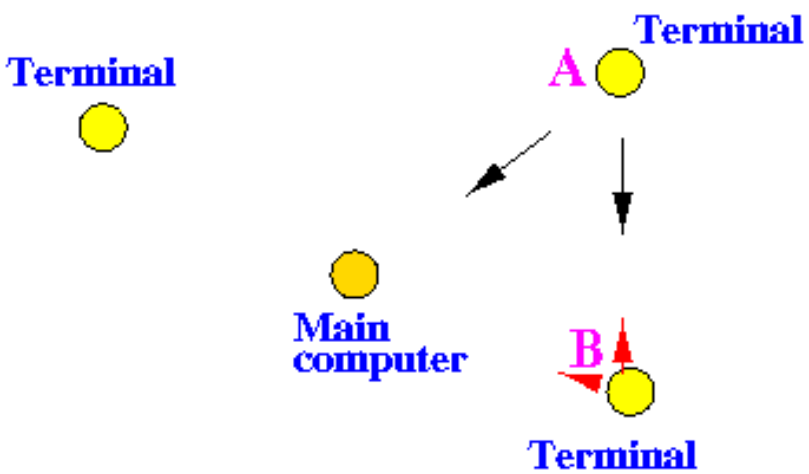
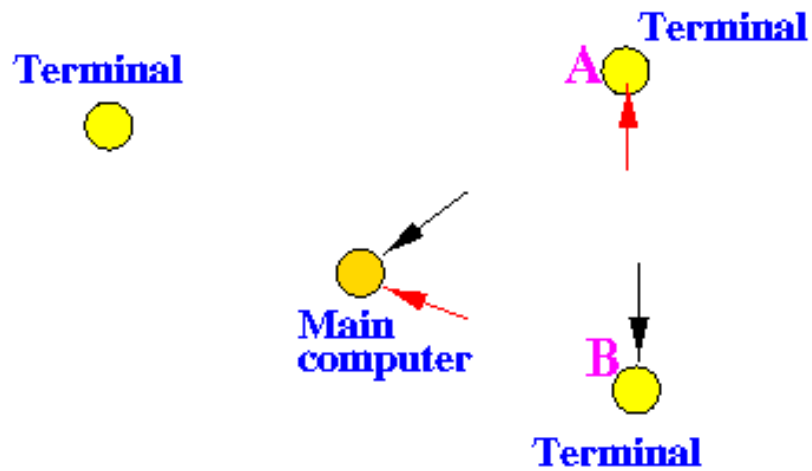


Node B senses that the channel is clear



But node B's transmission would collide with A's transmission at the main computer.....

Sequence of events:

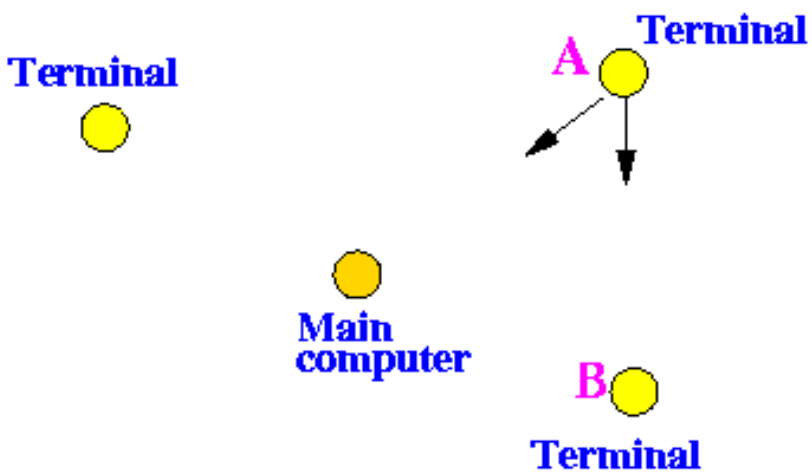
- Node **A starts a transmission**.
- The transmission **takes time** to reach node **B**
- Before **A**'s transmission reaches **B**, **B senses that the channel is idle**
- **Result:**
- transmissions from nodes **A** and **B** collide at the main computer even though **B** senses that the **channel was clear** when it started the transmissions....

Suppose a node senses the channel is *busy* (there is a transmission)

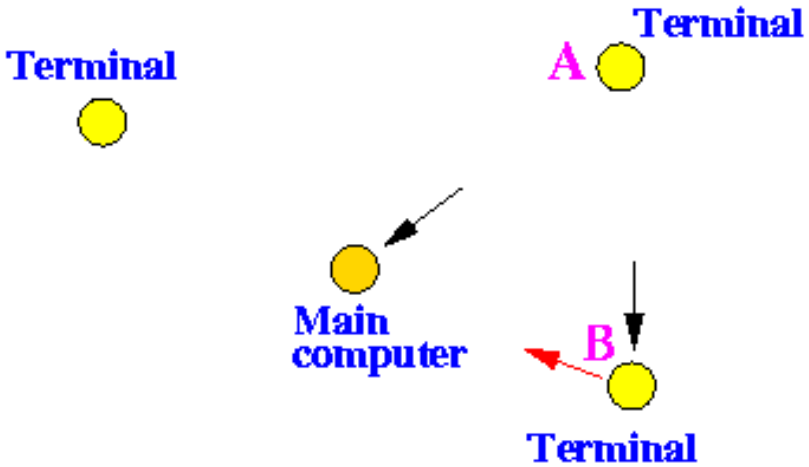
- **Question:**
- Can this **fact** help a node **reduce the likelihood** of a **collision** ?
- i.e., will a **collision certainly** occur when a **node** transmits ?

- **NO !!!**
- A node can get a **false channel busy** indication:
-

A node can get a **false channel busy** indication:



Node B senses that the channel is busy



But its transmission would NOT collide when it reaches the main computer !

Sequence of events:

- Node **A** starts transmitting a message.
- Due to the large distance, **A** will have finished transmitting the message, before the wave front of the message arrives at **B**
- When the wave front of **A**'s message arrives at **B**, node **B** wants to transmit a message.
- Node **B** senses the channel and detect **A**'s transmission.
- So with carrier sensing, **B** concludes that it should wait.

HOWEVER

- If **B** would transmit right away, the message of **B** would arrive at the main computer ***without suffering from collision*** !!
- That is because by the time that **B**'s message arrives at the main computer, **A**'s message has been cleared at the main computer (collision at the main computer occurs only when 2 transmission reach the main computer's antenna simultaneously)
- So **B** will get a **false indication of an busy channel**
- When **B** transmits while the channel is busy, its transmission will NOT collide with **A**'s at the main computer site !!!

Conclusion

- When you sense that the **channel is idle**; you **may** cause a **collision** if you **transmit**

- When you sense that the **channel is busy**;
  you **may *not*** cause a **collision** if you **do transmit**

In other words:

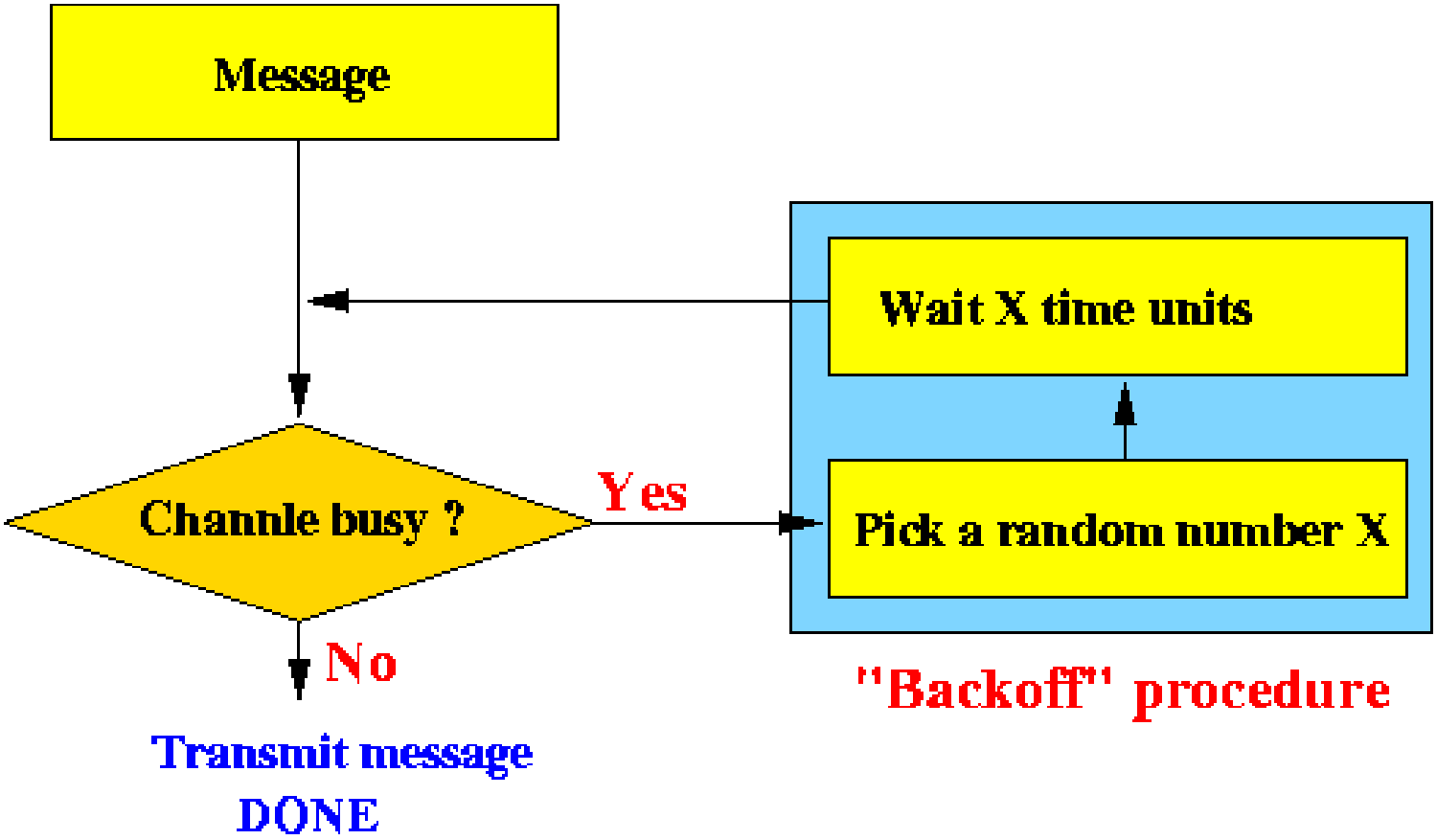
- you **did *not* learn anything (got any wiser)** by sensing the channel
- Put it bluntly:
- it's **useless** to sense the channel in ALOHA…

Different types of CSMA protocols

- There are **2 (or 3) types of CSMA protocols**:
- **Non-persistent CSMA** (the wimp)
- **1-persistent CSMA** (the stalker)
- (**General case: p-persistent CSMA** (the gambler))

Non-persistent CSMA:

- **The *non-persistent* CSMA: go away** when channel is **busy**
- If the node **senses** that the **carrier is busy**, it performs a **backoff procedure**:
- The node selects a **random number X**
- The node then **wait X time units (slots)** and repeat the protocol from the start....
- If the node **senses** that the **carrier is idle**, the node transmits in the **next slot**
- After transmitting a message, the CSMA protocol terminates.
  (Messages **do not** need to be ACKed.)


Message
Wait X time units
Channle busy ?
Yes
Pick a random number X
No
"Backoff" procedure
Transmit message
DONE

Design rationale of the non-persistent CSMA:

- The **non-persistent CSMA** is a **"pessimistic" protocol**:

- "I senses the channel is busy...**Pessimist:** I believe that the system is **heavily loaded**
- I.e., there are **other nodes** out there that are **also trying to transmit**
- So I better **come back later** (The **random wait** will **minimize** the likelihood that **2 or more nodes** will start transmission at the same time...)

Strength of *non-persistent CSMA*

- Achieves **good performance** in **heavily loaded systems** - when there are many nodes that have messages to transmit (heavily loaded), they will "auto schedule" themselves with the random

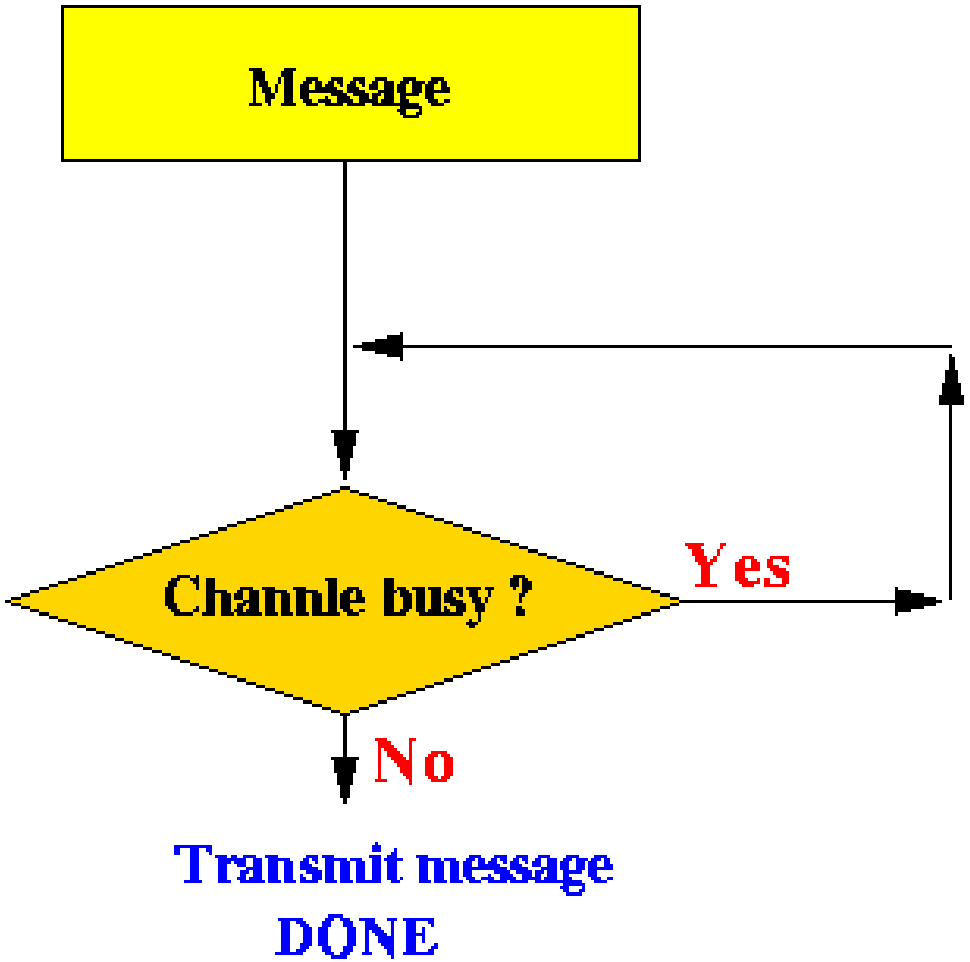
Weakness of *non-persistent CSMA*

- **Throughput** (# successful transmissions per time unit) can be **low** in a **light load system**
- E.g.: if there is **only one node waiting**, it can **safely transmit as soon as** the **current transmission** ***terminates***

1-persistent CSMA:

- The **1-persistent CSMA** protocol: **stalk** with a **purpose**....
- If the **node senses** that the **carrier** is **busy**, it **wait (stalks)** until the channel becomes **idle**
- If the **node senses** that the **carrier** is **idle**, the node transmits the **message** in the **next slot**
- After transmitting a message, the CSMA protocol terminates.
  (Messages **do not** need to be ACKed.)
-

Schematically:



NOTE:

- When **2 or more nodes** are **waiting** for the ***current*** **transmission** to end (using 1-persistent CSMA), their **transmissions will *surely* collide** when the current transmission ends...

Rationale of the 1-persistent CSMA:

- **1-persistent CSMA** is an **"optimistic" protocol**

- "I sense the channel is busy, and I believe that the system is **lightly loaded**(I.e., I'm the **only one** out there that is waiting to transmit)....

- So I'll just wait until this fellow is done and then take my turn"

Strength of the *1-persistent* CSMA protocol:

- Performs very well in **lightly loaded systems**- i.e., when there are **few nodes** that have messages to transmit

- A node that finds the channel busy will most likely be the only node waiting for the channel to become clear and it will start transmitting as soon as the current transmission is done and will not **waste any time**.

Strength of the *1-persistent* CSMA protocol:

- **Performance** will **suck at heavy load**
- There will be an **excessive number** of **collisions**.

p-persistent CSMA:

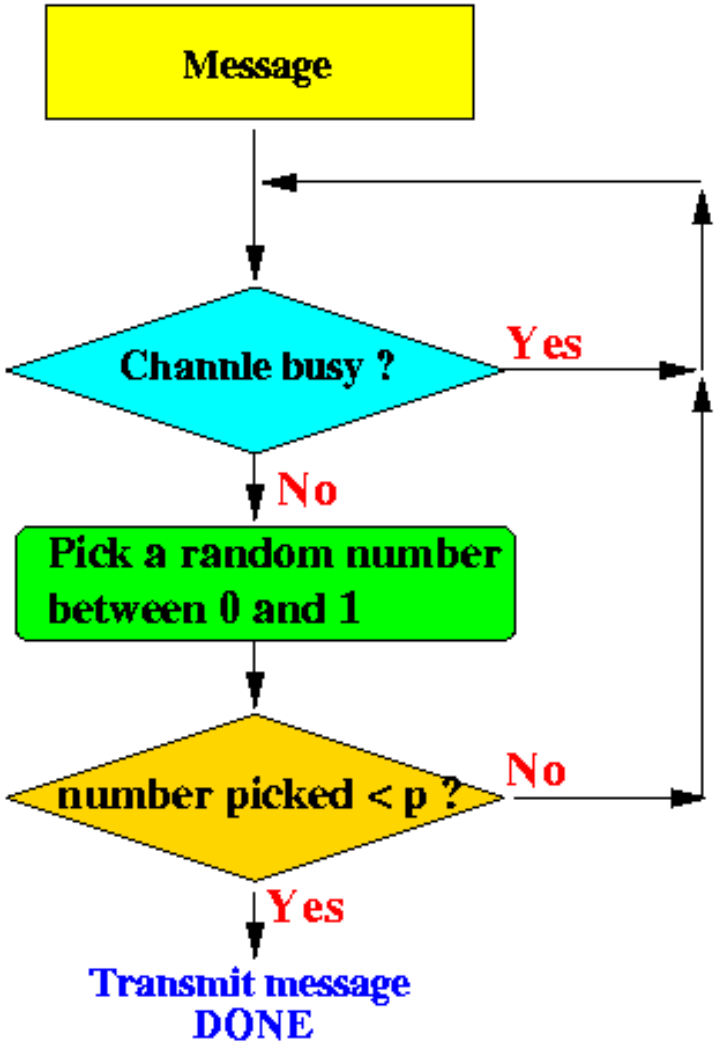
- The **p-persistent CSMA** protocol: the **gambler** (middle ground between the **non-persistent** and **1-persistent** CSMA)
- If the **node senses** that the **carrier** is **busy**, it **wait (stalks)** until the channel becomes **idle**

- If the **node senses** that the **carrier** is **idle**:

- The **node flip a coin** (pick a **random number**)
- If **coin comes up head** (**random number < p**), transmits the **message** in the **next slot**
- Otherwise, **try again** from the start....

Schematically:

- After transmitting a message, the CSMA protocol terminates.
  (Message **do not** need to be ACKed.)



NOTE:

- When **2 or more nodes** are **waiting** for the ***current*** **transmission to end** (using ***p*-persistent CSMA**, they will **only collide** when:
- ***2 or more*** **nodes** pick a **random number** that is **less than *p***

- **Rationale of the p-persistent CSMA:**
- The **p-persistent CSMA** is **designed** to **avoid** the **collision problem** in **1-persistent CSMA**

- By **picking a random number**, the **system** tries to ensure that **at most 1 node** will **transmit**
- **What is the "right value" for *p* ?**
-

What is the "right value" for *p* ?

- *p* must be set to a value so that it is **highly likely** that **exactly one node** will pick a **random number** that is **less than *p***

- **For example**, if **on the average**, there are **2 nodes** that are waiting for a transmission to finish, then:

- set ***p*** = 1/2 = 0.5.

- **On the other hand**, if **on the average**, there are **3 nodes** that are waiting for a transmission to finish, then:

- set ***p*** = 1/3 = 0.3333

- And so on...

Conclusion:

p must be **dynamic**

- It's **hard** to **estimate the number of nodes** that is **busy**...

- So the ***p*-persistent CSMA** protocol is **not implemented** as a real life protocol.
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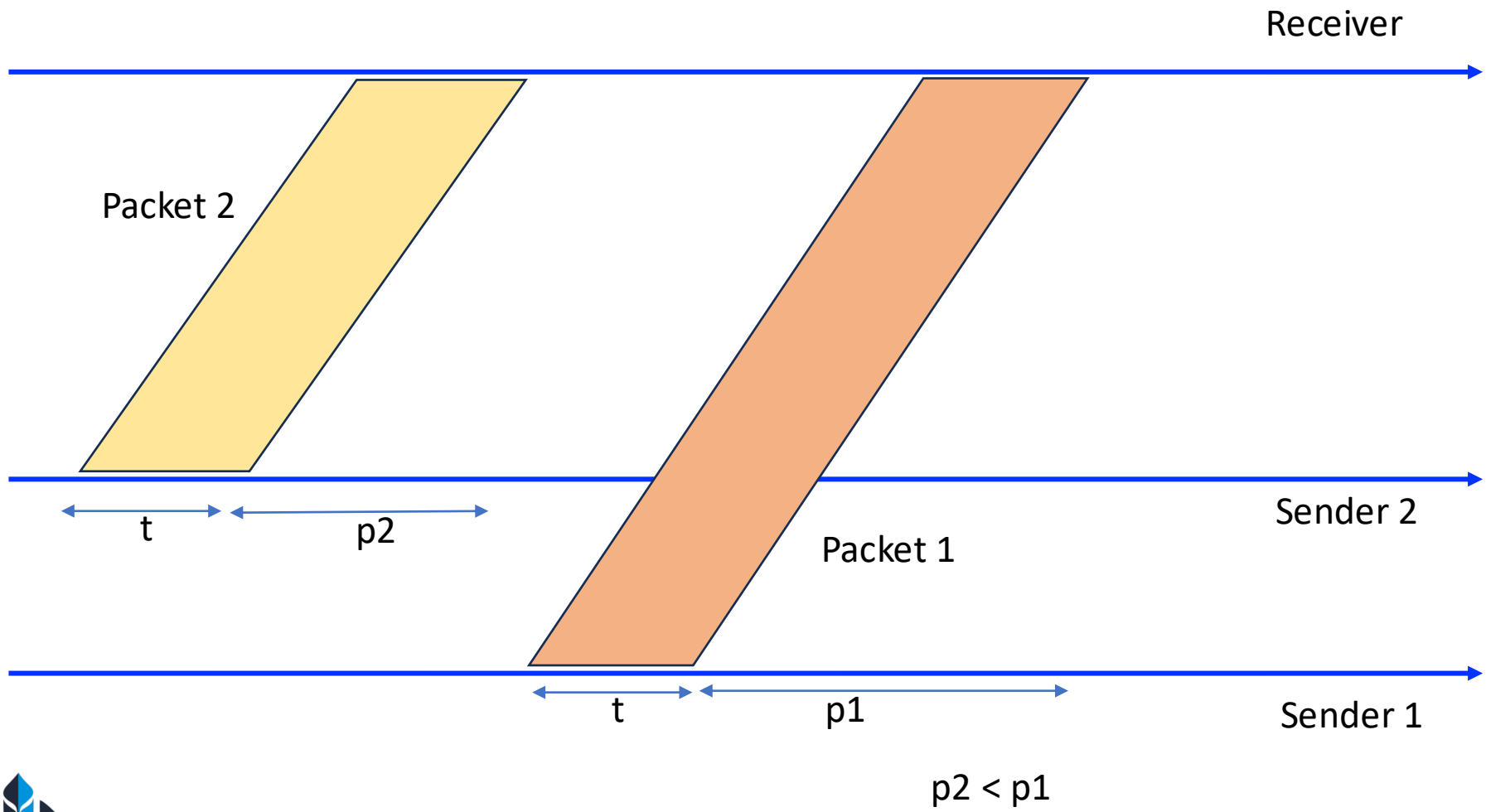
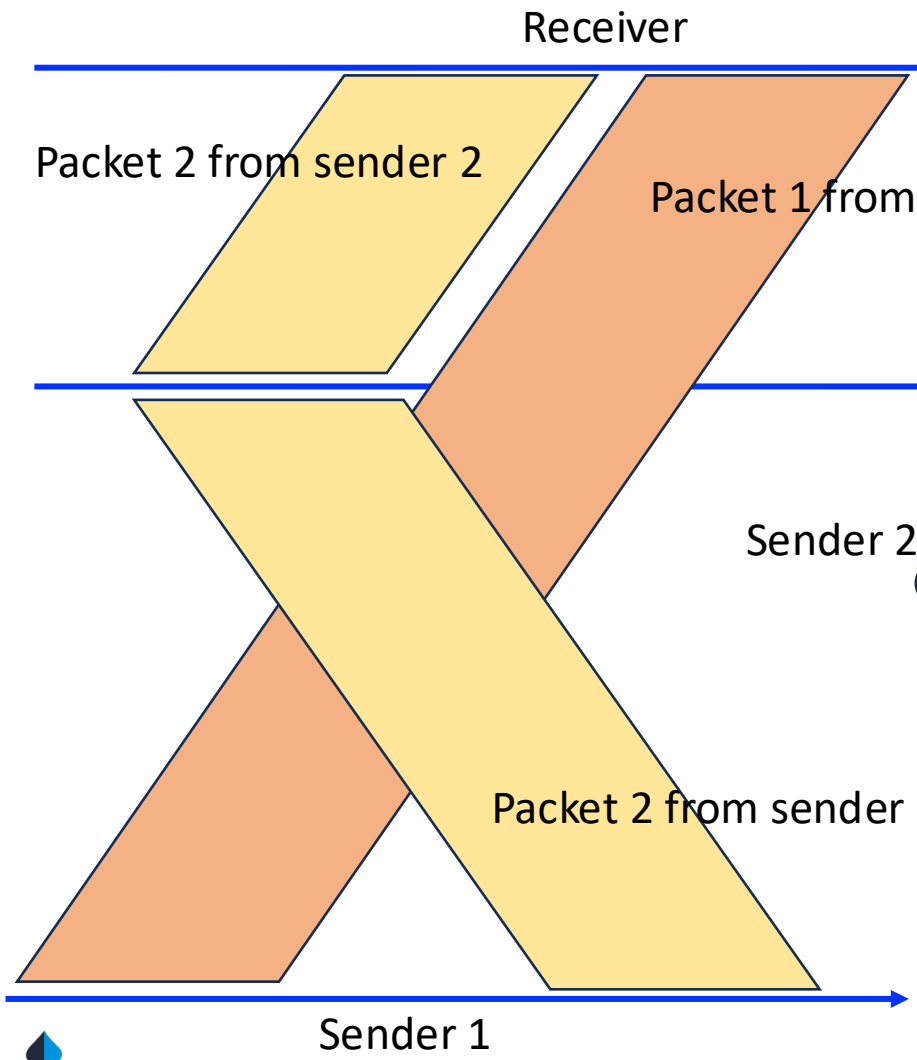
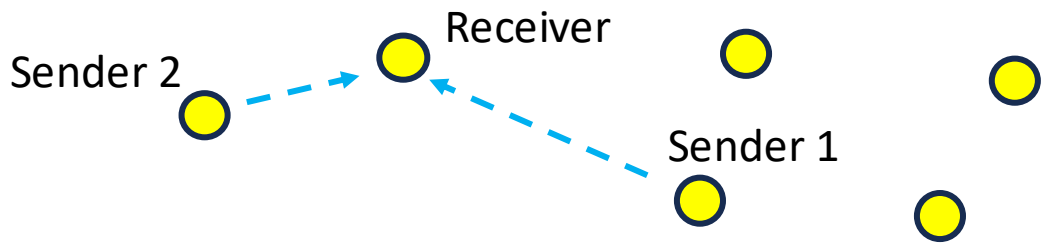
Aloha Setting

- No collision



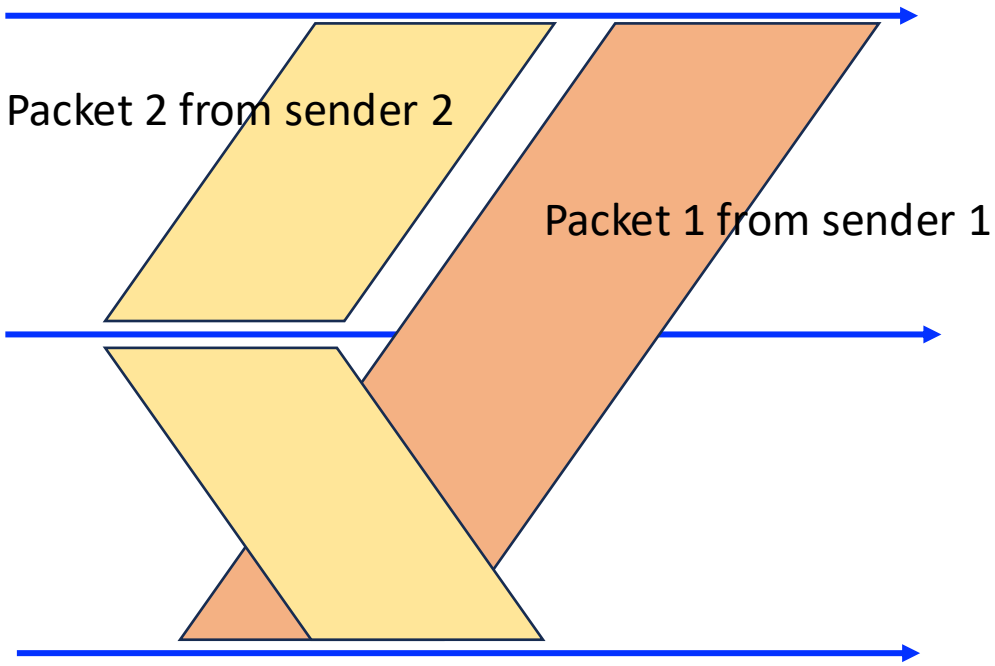
Scenario when distance is high

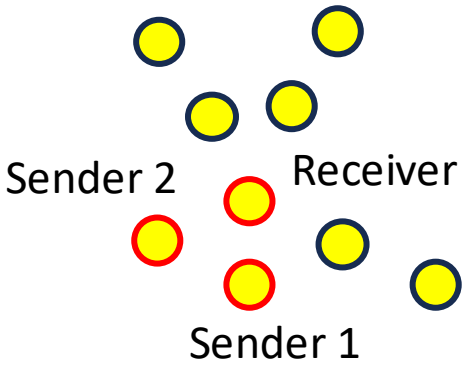




Sender 1 senses channel, finds no signal, sends

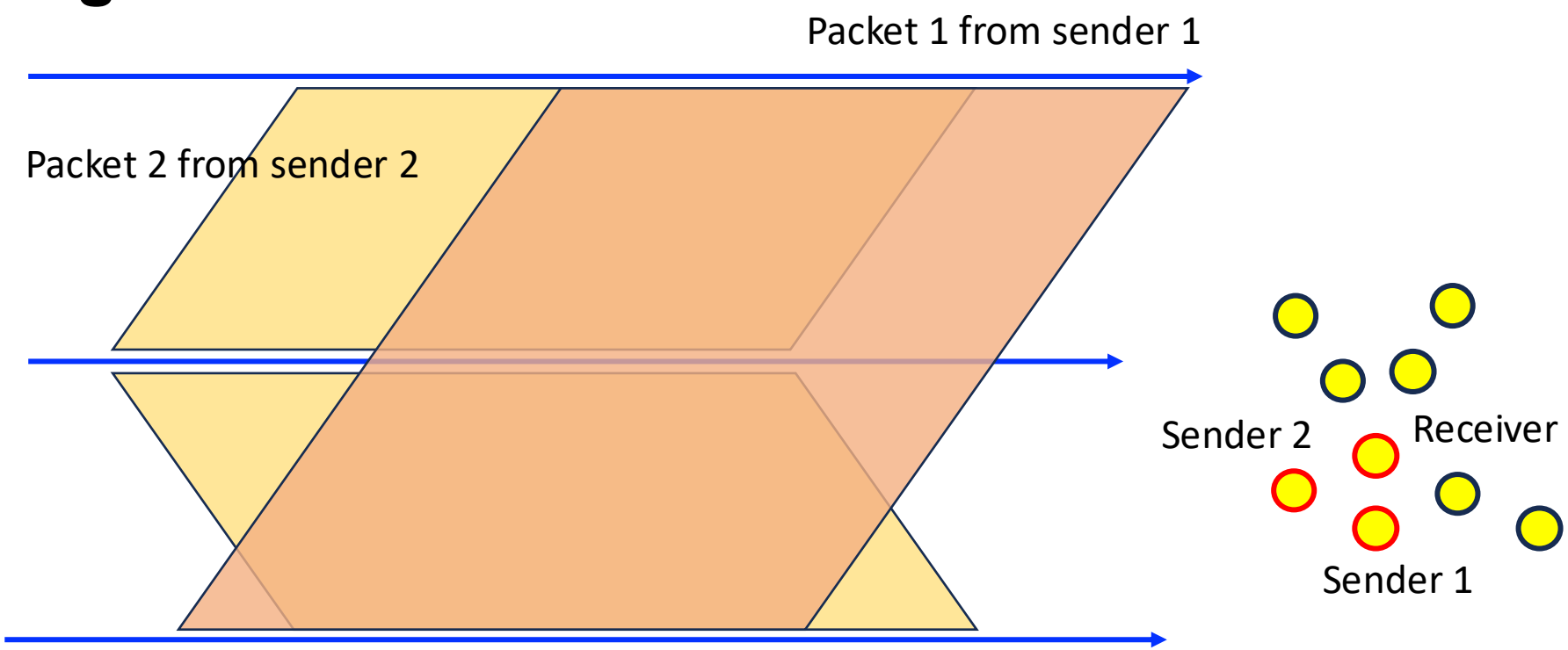
When distance is low



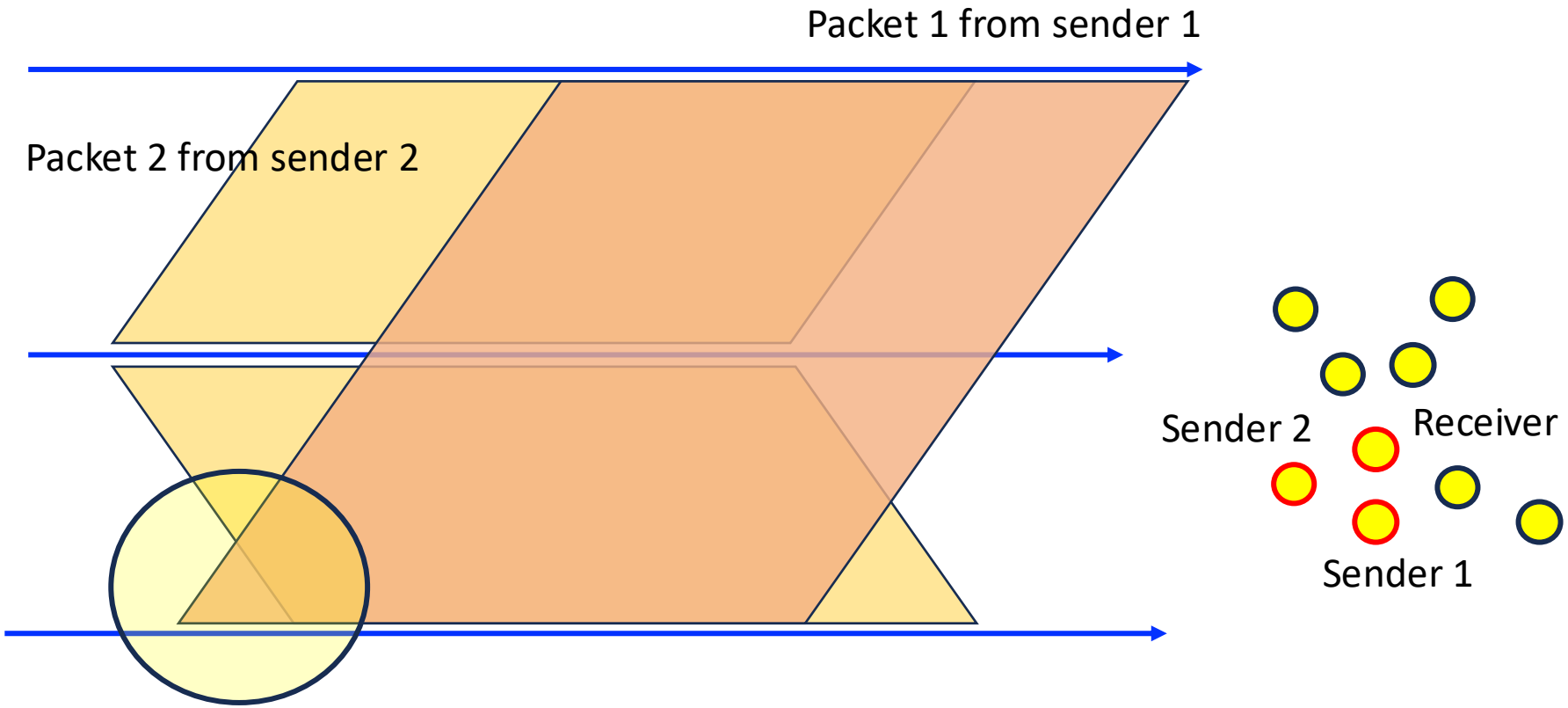


Sender 1 keeps sending even if sensing results positive, But still no collision? Why?

When distance is low but transmission time is higher



When distance is low but transmission time is higher



When packet length is larger than propagation Collision is more likely – when we transmit even if channel busy